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User Guide



Applies to version 3.9 of the timing capture software and version 3.3 of the registration database.

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1. Overview

1.1. Introduction

RATS includes two components:

- A registration database built in MS Access 2016 (hereafter called RATSdb)
- Timing supervision and data capture software built in MS VisualStudio in VB (hereafter called RATSts)

1.2. Registration Database - RATSdb

The screenshot shows the 'Context' window of the Registration And Timing System (RATS). The window has a red title bar and a menu bar with options: File, Home, Create, External Data, Database Tools, Help, and a search icon. The main area is titled 'Context' and contains several input fields and buttons. The 'Event ID' field is set to 17, and the 'Effective Date' is 30-Apr-18. The 'Current Event' field is 'Techfest 2018 Rehearsal'. There are buttons for 'New Event', 'Refresh', 'Record Competition Entries', 'List Competitions', 'List Entries', 'Manual Input', 'Name Scores', and 'Scores Report'. A 'Navigation Pane' is visible on the left. The status bar at the bottom shows 'Record: 14', '1 of 9', 'No Filter', and 'Search'. The database version is 2.1.

1.3. Timing Supervisor - RATSts

The screenshot shows the 'Registration And Timing System (RATS) Version 3.9.1 database version 3.3' window. The title bar is grey. The main area is titled '17/03/2024 UKMARS Spring 2024' and 'Practice Mode'. It displays a table with columns: Course, To Go, Run No, of, Touches, BestScore, Rank, Last Run, LastScore, and SplitTime. The table shows a single row with values: 0.00, To Go, 0, no, 0, 0, 0, 0.000, 0.000, 0.00. Below the table, there are sections for 'Timing Gates State' (Waiting start present), 'Touch Enabled' (Touch Enabled), 'Touch Divider' (Touch Divider), 'Touch Cumtive' (Touch Cum), 'Touch Time' (Touch Time), 'Course Divider' (Course Divider), and 'Maze Divider' (Maze Divider). There are also buttons for 'Switch Mode', 'Practice <=> Contest', 'Entry Controls', 'Add Touch', 'DNF', 'Clear Display', 'New Robot', '1 Channel Timer', '800x600', '1280x720', 'Results', '612 x 595', 'Run Order', '2 Channel Timer', 'Display', and 'Results'. The status bar at the bottom shows 'NoMont', 'Verbose', 'Clear', and 'Calibrate'.

2. Registration Database - RATSdb

2.1. The CONTEXT or start-up window

On launch, the RATSdb displays the context form:

The screenshot shows the 'Context' form in the 'Registration And Timing System (RATS)' application. The form is titled 'Context' and contains several input fields and buttons. Annotations with orange callouts highlight specific features:

- A callout points to the 'New Event' button, stating: "If you need to setup a new event – click this button".
- A callout points to the 'Refresh' button, stating: "Refresh button ensures all events are visible".
- A callout points to the 'Record Entries' button, stating: "You can step through Robotics Events already setup using record navigation".

The form includes the following fields and buttons:

- Event ID: 36
- Effective Date: 17/03/2024
- Current Event: UKMARS Spring 2024
- Buttons: New Event, Refresh, Record Entries, Sequence Entries, List Contests, List Entries, Manual Input, Scores Report, Unsplit Scores, All Runs, Export Entries, Export Results, Export Markdown.
- Footer: Front End Version 3, Database Version, Record: 1 of 20, No Filter, Search, Num Lock.

The context form remains throughout the RATSdb session and is used to establish which Robotics Event is being managed and launch child windows to allow registration and entry of Robots into competitions.

An Entity Relationship model for RATSdb is included as Appendix A.

If your event has not already been setup click on **NEW EVENT** to create it.

This principle of operation is used throughout RATSdb. Normally you will be choosing from lists of values already setup, but if the value needed does not already exist then a button will be nearby to create the new value. Eg a new mouse, a new contestant, a new competition etc

2.2. Creating a New Robotics Event

Registration And Timing System (RATS) Sign in

File Home Create External Data Database Tools Help Tell me what you want to do

Context

Robotics_Event

New Robotics Event

Event Name: Another Event

Event Date: May 2024

Existing Events

	Mo	Tu	We	Th	Fr	Sa	Su	
UKMARS Spring 20	29	30	1	2	3	4	5	Mar-24
UKMARS Autumn 2	6	7	8	9	10	11	12	Oct-23
UKMARS National	13	14	15	16	17	18	19	Jul-23
UKMARS Spring 20	20	21	22	23	24	25	26	Mar-23
UKMARS Autumn 2	27	28	29	30	31	1	2	Oct-22
UKMARS TechFest	3	4	5	6	7	8	9	Jun-22
InnovationFest 2022								25-May-22
Minos 2022								03-Apr-22
Version 3.9 Testing								05-Feb-22
UKMARS Autumn 2021								02-Oct-21
Version 3.7 Testing								06-Nov-19
UKMARS Autumn 2019								12-Oct-19
UKMARS Techfest 2019								01-Jun-19
Testing for Techfest 2019								03-May-19
Minos 2019								07-Apr-19

Today

Front End Version 3.7
Database Version 3.3b

Record: 1 of 20 No Filter Search

Form View Num Lock

2. Enter the name of your new event here

4. When you've set the name and date just close the form to save the event

3. Set the date for your new event with the date

1. The sub-form loads showing all the events that are already defined

The new Robotics Event sub-form is presented in front of the Context Form. When done close the form and return to the Context Form.

2.3. Registering New Entries

The screenshot shows the 'Context' window of the Registration And Timing System (RATS). The window has a red title bar with the text 'Registration And Timing System (RATS)' and a 'Sign in' button. Below the title bar is a menu bar with 'File', 'Home', 'Create', 'External Data', 'Database Tools', 'Help', and 'Tell me what you want to do'. The main content area is titled 'Context' and contains several input fields and buttons. The 'Event ID' field is set to '36', the 'Effective Date' field is set to '17/03/2024', and the 'Current Event' field is set to 'UKMARS Spring 2024'. There are buttons for 'New Event', 'Refresh', 'Record Entries', 'Sequence Entries', 'List Contests', 'List Entries', 'Manual Input', 'Scores Report', 'Unsplit Scores', 'All Runs', 'Export Entries', 'Export Results', and 'Export Markdown'. The 'Record Entries' button is highlighted with an orange circle and a callout that says '2. Click this button to start registering robots'. Another callout points to the 'Current Event' field, saying '1. Make sure you have the right Robotics Event'. The bottom of the window shows 'Front End Version 3.7', 'Database Version 3.3b', and a status bar with 'Record: 1 of 20', 'No Filter', and a 'Search' button.

1. Make sure you have the right Robotics Event

2. Click this button to start registering robots

Registration And Timing System (RATS)

File Home Create External Data Database Tools Help Tell me what you want to do

Context Entry Form

UKMARS Spring 2024

Robot Name: Asymouse Contestant Name: Ian Butterworth Contest Entry: 0

New Robot New Contest Enter Robot

Today's entries for this robot

Entry ID	Robot Name	Contest Name	Contest Class	Contest Challenge
1217	Asymouse	Senior Line Follower	Final	Line Follower
1218	Asymouse	Senior Wall Follower	Final	Non Contact Wall Follower
1219	Asymouse	Maze Solver	Final	Maze Solver

List Entries Close Form

Form View Num Lock

2. Click this button to get a list of competitions for the current event to create a new entry

1. A list of entries already entered for this robot for this robotics event is displayed – check you are not creating a duplicate

Registration And Timing System (RATS)

File Home Create External Data Database Tools Help Tell me what you want to do

Context Entry Form

UKMARS Spring 2024 17-Mar-24

Robot Name: Asymouse Contestant Name: Ian Butterworth Contest Entry: 0 Where Registered: OnLineForm

New Robot New Contest Enter Robot

Today's entries for this robot

Entry ID	Robot Name	Contest Name	Contest Class	Contest Challenge
1217	Asymouse	Senior Line Follower	Final	Line Follower
1218	Asymouse	Senior Wall Follower	Final	Non Contact Wall Follower
1219	Asymouse	Maze Solver	Final	Maze Solver

List Entries Close Form

Form View Num Lock

1. Select the competition you want to enter this robot for

Registration And Timing System (RATS) Sign in

File Home Create External Data Database Tools Help Tell me what you want to do

Context Entry Form

UKMARS Spring 2024 17-Mar-24

Robot Name: Asymouse Contestant Name: Ian Butterworth Contest Entry: 94 Half Size Maze Solver Final Where Registered: OnLineForm

New Robot New Contest

Today's entries for this robot

Entry ID	Robot Name	Contest Name	Contest Class	Contest Challenge
1217	Asymouse	Senior Line Follower	Final	Line Follower
1218	Asymouse	Senior Wall Follower	Final	Non Contact Wall Follower
1219	Asymouse	Maze Solver	Final	Maze Solver

List Entries Close Form

Form View Num Lock

1. Use this drop-down list to record where the registration request was received. Defaults to OnLineForm

Registration And Timing System (RATS) Sign in

File Home Create External Data Database Tools Help Tell me what you want to do

Context Entry Form

UKMARS Spring 2024 17-Mar-24

Robot Name: Asymouse Contestant Name: Ian Butterworth Contest Entry: 94 Half Size Maze Solver Final Where Registered: OnLineForm

New Robot New Contest

Today's entries for this robot

Entry ID	Robot Name	Contest Name	Contest Class	Contest Challenge
1217	Asymouse	Senior Line Follower	Final	Line Follower
1218	Asymouse	Senior Wall Follower	Final	Non Contact Wall Follower
1219	Asymouse	Maze Solver	Final	Maze Solver

List Entries Close Form

Form View Num Lock

2. You can click this button to see a complete list of entries for this Robotics Event

1. Click this button to enter this robot

2.4. Creating a New Robot Record

Registration And Timing System (RATS)

File Home Create External Data Database Tools Help Tell me what you want to do

Sign in

Context Entry Form

UKMARS Spring

Robot Name

New Robot

Today's entries for the

Contestant ID

Contestant Name

Further Information

List Entries

Form View

Num Lock

1. If this is the 1st entry for this Robot Click this button to open the New Robot Form

2. Choose from the drop-down list of

Contestant ID	Contestant Name	Contestant Level
1	Ian Butterworth	Senior
31	Ian Buxton	Senior
117	Ian Robinson	Senior
58	Ilsa Adnan	Junior
60	Imogen Crocker	Junior
62	India Harris	Junior
121	Jamie Coombs	Senior
14	Jim Chidley	Senior
110	Joe Smith	Junior
73	John Fisher	Senior
103	Julian	Junior
113	Julian Lenczewski	Junior
129	Julian S Lenczewski	Junior
21	Junior Test	Junior
96	Kaviraj A	Student
36	Kaviyashree Damodaran	Junior

3. Enter the name of the New Robot here

Registration And Timing System (RATS)

File Home Create External Data Database Tools Help Tell me what you want to do

Sign in

Context Entry Form

UKMARS Spring

Robot Name

New Robot

Today's entries for the

Contestant ID

Contestant Name

Further Information

List Entries

Form View

Num Lock

3. Enter the name of the New Robot here

4. When you've set the name and Contestant just close the form to save the Robot

5. If this is the 1st entry for this Contestant Click this button to open the New Contestant Form

2.5. Creating a New Contestant Record

The screenshot displays the 'Registration And Timing System' interface. A 'New Contestant' form is open, overlaid on a background window showing 'Today's entries for this' with a table of 'Entry ID' and '(New)'. The form has fields for 'Contestant Name', 'Contestant Class' (a dropdown menu with options: Junior, Intermediate, Senior), and 'Email Address'. Four numbered callouts provide instructions: 1. Enter the name of the New Contestant here (points to the Contestant Name field); 2. Choose Class of the Contestant here (points to the Contestant Class dropdown); 3. Enter the email address of the New Contestant here (points to the Email Address field); 4. When you've set the name class and email address just close the form to save the Contestant (points to the close button in the top right corner of the form). The interface also includes a 'Robot Name' dropdown, a 'New Robot' button, and a 'List Entries' button at the bottom.

1. Enter the name of the New Contestant here

2. Choose Class of the Contestant here

3. Enter the email address of the New Contestant here

4. When you've set the name class and email address just close the form to save the Contestant

2.6. Creating a New Contest

Registration And Timing System (RATS) Sign in

File Home Create External Data Database Tools Help Tell me what you want to do

Context Entry Form

UKMARS Spring 2024 17-Mar-24

Robot Name Contestant Name Contest Entry Where Registered

New Robot New Contest Enter Robot

New Contest

Event ID 36

Event Name UKMARS Spring 2024

Event Date 17-Mar-24

Challenge

Contest Name

Contest Class

Contest Date 17-Mar-24

Scoring Model

Entry Time Limit Secs 600

Grace Period Secs 0

Runs Allowed 5

Today's Contests

Challenge	Event ID	Contest Class
Senior Line Follower	36 Final	
Senior Half Size Line Follower	36 Final	
Senior Wall Follower	36 Final	
Drag Race	36 Final	
Half Size Wall Follower	36 Final	
Half Size Maze Solver	36 Final	
Maze Solver	36 Final	
Junior Half Size Line Follower	36 Final	
Junior Line Follower	36 Final	
Junior Wall Follower	36 Final	

1. If the Contest you want to enter has not been set up, click on this button to open the New Contest Form

2. A list of Contests already entered for this robotics event is displayed – check you are not creating a duplicate

New Contest

New Contest

Event ID

36

Event Name

UKMARS Spring 2024

Event Date

17-Mar-24

Challenge

Any Challenge Free Practice

Drag Race

Half Size Line Follower

Half Size Maze Solver

Half Size Wall Follower

Line Follower

Maze Solver

Non Contact Wall Follower

Time Trial

New Challenge

1. Select the Challenge for this Competition

Today's Contests

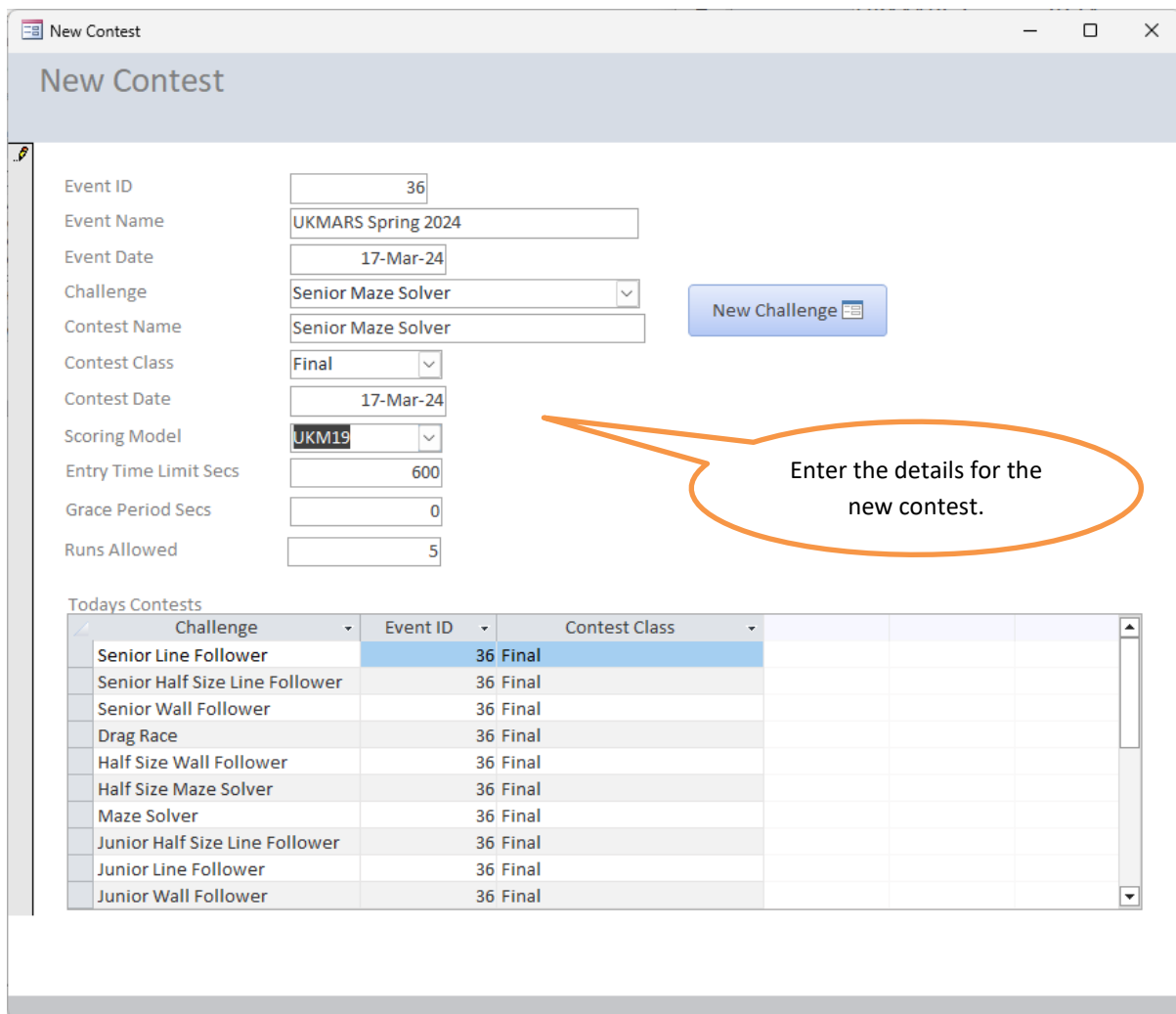
Challenge	Event ID	Contest Class
Senior Line Follower	36 Final	
Senior Half Size Line Follower	36 Final	
Senior Wall Follower	36 Final	
Drag Race	36 Final	
Half Size Wall Follower	36 Final	
Half Size Maze Solver	36 Final	
Maze Solver	36 Final	
Junior Half Size Line Follower	36 Final	
Junior Line Follower	36 Final	
Junior Wall Follower	36 Final	

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Document version 2.0

May 2024



New Contest

Event ID: 36

Event Name: UKMARS Spring 2024

Event Date: 17-Mar-24

Challenge: Senior Maze Solver

Contest Name: Senior Maze Solver

Contest Class: Final

Contest Date: 17-Mar-24

Scoring Model: UKM19

Entry Time Limit Secs: 600

Grace Period Secs: 0

Runs Allowed: 5

Today's Contests

Challenge	Event ID	Contest Class
Senior Line Follower	36	Final
Senior Half Size Line Follower	36	Final
Senior Wall Follower	36	Final
Drag Race	36	Final
Half Size Wall Follower	36	Final
Half Size Maze Solver	36	Final
Maze Solver	36	Final
Junior Half Size Line Follower	36	Final
Junior Line Follower	36	Final
Junior Wall Follower	36	Final

Enter the details for the new contest.

Contest name defaults to the selected challenge and can then be edited

Contest class selected from drop down - either Final or Heats

Scoring model selected from drop down - determines how entries are scored :

- UKM19 UK maze solver 2019 rule set
- UKWF19 UK wall follower 2019 rule set

Entry time specifies the number of seconds in total a robot entry is allowed

Grace period specifies the number of seconds a robot is allowed additional to the entry time to complete a run that commenced prior to the end of the run time

2.7. Creating a New Challenge

The 'New Contest' form contains the following fields:

- Event ID: 36
- Event Name: UKMARS Spring 2024
- Event Date: 17-Mar-24
- Challenge: (dropdown menu)
- Contest Name: (text input)
- Contest Class: (dropdown menu)
- Contest Date: 17-Mar-24
- Scoring Model: (dropdown menu)
- Entry Time Limit Secs: 600
- Grace Period Secs: 0
- Runs Allowed: 5

A blue button labeled 'New Challenge' is located to the right of the 'Challenge' dropdown. A callout bubble points to this button with the text: "1. If the Challenge you want to enter has not been set up, click on this button to open the New Challenge Form".

Below the form is a table titled 'Todays Contests':

Challenge	Event ID	Contest Class
Senior Line Follower	36 Final	
Senior Half Size Line Follower	36 Final	
Senior Wall Follower	36 Final	
Drag Race	36 Final	
Half Size Wall Follower	36 Final	
Half Size Maze Solver	36 Final	
Maze Solver	36 Final	
Junior Half Size Line Follower	36 Final	
Junior Line Follower	36 Final	
Junior Wall Follower	36 Final	

The 'New Challenge' form contains the following sections:

- Challenge Type:** A text input field. A callout bubble points to it with the text: "2. Enter the description of the New Challenge here".
- Existing Challenges:** A list of challenges already entered for this event. A callout bubble points to the list with the text: "1. A list of Contests already entered for this robotics event is displayed – check you are not creating a duplicate".

The 'Existing Challenges' list includes:

- Half Size Maze Solver
- Half Size Wall Follower
- Line Follower
- Maze Solver
- Non Contact Wall Follower

2.8. Sequencing the Running Order of Entries

The screenshot displays the 'Context' page of the Registration And Timing System (RATS). The interface includes a top navigation bar with a 'Sign in' button and a menu with options like File, Home, Create, External Data, Database Tools, and Help. A left-hand 'Navigation Pane' is visible. The main content area is titled 'Context' and contains several input fields and buttons. The 'Event ID' field is set to 36, 'Effective Date' is 17/03/2024, and 'Current Event' is UKMARS Spring 2024. Buttons for 'New Event', 'Refresh', 'Record Entries', 'Sequence Entries', 'List Contests', 'List Entries', 'Manual Input', 'Scores Report', 'Unsplit Scores', 'All Runs', 'Export Entries', 'Export Results', and 'Export Markdown' are present. A callout bubble points to the 'Sequence Entries' button with the text 'Click here to open the sequencing function'. The bottom of the page shows version information (Front End Version 3.7, Database Version 3.3b) and a status bar with 'Record: 1 of 20', 'No Filter', and a 'Search' field.

The order in which entries are presented for a contest can be determined by entering ascending values for a sequencing number for a contest. If no sequence numbers are assigned all entries will default to the highest sequence number and the entries will be presented in the order that they were registered on the database.

Registration And Timing System (RATS) Form Tools Sign in

File Home Create External Data Database Tools Help Form Datasheet Tell me what you want to do

Context Today's Competitions

Sequence Competitions

Navigation Pane

Competition_Name Senior Wall Follower

Competition_Class Final

Entry

Mouse_Name	Sequence_Number
Freda	10
WOW	20
Monty Pi	30
mwmouse	40
Pywall	50
Magellan	60
StarFire	70
Half Monty	80
Erratic	90
Orange	100
Asymouse	110
MontyMouse	120

Record: 1 of 12 No Filter Search

Form View Num Lock

2. Enter sequence numbers to control the order in which entries will run in the contest

1. Navigate to the Contest whose entries you want o sequence

2.9. Registration Reports

The screenshot shows the 'Context' form in the RATS application. The form has a red header bar with the title 'Registration And Timing System (RATS)' and a 'Sign in' button. Below the header is a navigation pane on the left with a 'Context' tab. The main area contains several input fields and buttons. The 'Event ID' field is set to '36', the 'Effective Date' is '17/03/2024', and the 'Current Event' is 'UKMARS Spring 2024'. There are buttons for 'New Event', 'Refresh', 'Record Entries', 'List Entries', 'Manual Input', 'Scores Report', 'Unsplit Scores', 'All Runs', 'Export Entries', 'Export Results', and 'Export Markdown'. Two callouts are present: one pointing to the 'List Contests' button with the text '1. Launch report to list the Contests for this Event.', and another pointing to the 'List Entries' button with the text '2. Launch report to list the Entries for all the Contests for this Event.' The bottom of the form shows 'Front End Version 3.7', 'Database Version 3.3b', and a status bar with 'Record: 1 of 20', 'No Filter', and a 'Search' button.

Registration And Timing System (RATS) Sign in

File Home Create External Data Database Tools Help Tell me what you want to do

Context

Event ID 36 New Event

Effective Date 17/03/2024 Refresh

Current Event UKMARS Spring 2024

Record Entries List Entries

Manual Input

Scores Report Unsplit Scores All Runs

Export Entries Export Results Export Markdown

Front End Version 3.7
Database Version 3.3b

Record: 1 of 20 No Filter Search

Form View Num Lock

1. Launch report to list the Contests for this Event.

2. Launch report to list the Entries for all the Contests for this Event.

Two reports are available from the Context form to allow the competitions and the entries for the Robotics Event to be confirmed.

2.10. List Competitions

Registration And Timing System (RATS) Sign in ? - X

File Home Create External Data Database Tools Help Tell me what you want to do

Context Competitions_for_current_Event

Competitions for the current Event Tuesday, April 17, 2018 9:41:50 AM

Comp ID	Event	Challenge	Class	Competition Date	Scoring Model
41	17	Line Follower	Heats	30-Apr-18	UKLF
40	17	Non Contact Wall Follower	Heats	30-Apr-18	UKNCWF
43	17	Non Contact Wall Follower	Final	30-Apr-18	UKNCWF
42	17	Line Follower	Final	30-Apr-18	UKLF

4

Page 1 of 1

Report View Num Lock

2.11. List Entries

Registration And Timing System (RATS) Sign in ? - X

File Home Create External Data Database Tools Help Tell me what you want to do

Context Entries_for_current_Event

Entries for the current Event Tuesday, April 17, 2018 9:43:22 AM

Entry	Mouse Name	Comp	Used	Challenge	Class
171	Orange	41	☐	Line Follower	Heats
169	Asymouse	41	☐	Line Follower	Heats
2					
168	Freda	40	☒	Non Contact Wall Follower	Heats
167	Asymouse	40	☒	Non Contact Wall Follower	Heats
2					
172	Orange	42	☐	Line Follower	Final
170	Asymouse	42	☐	Line Follower	Final
2					
174	Asymouse	43	☐	Non Contact Wall Follower	Final
1					
7					

Page 1 of 1

Report View Num Lock

2.12. Manual Timing Capture

Registration And Timing System (RATS) Sign in

File Home Create External Data Database Tools Help Tell me what you want to do

Context

Navigation Pane

Context

Event ID 36 New Event

Effective Date 17/03/2024 Refresh

Current Event UKMARS Spring 2024

Record Entries Sequence Entries

List Contests List Entries

Manual Input

Scores Report Unsplit Scores All Runs

Export Entries Export Results Export Markdown

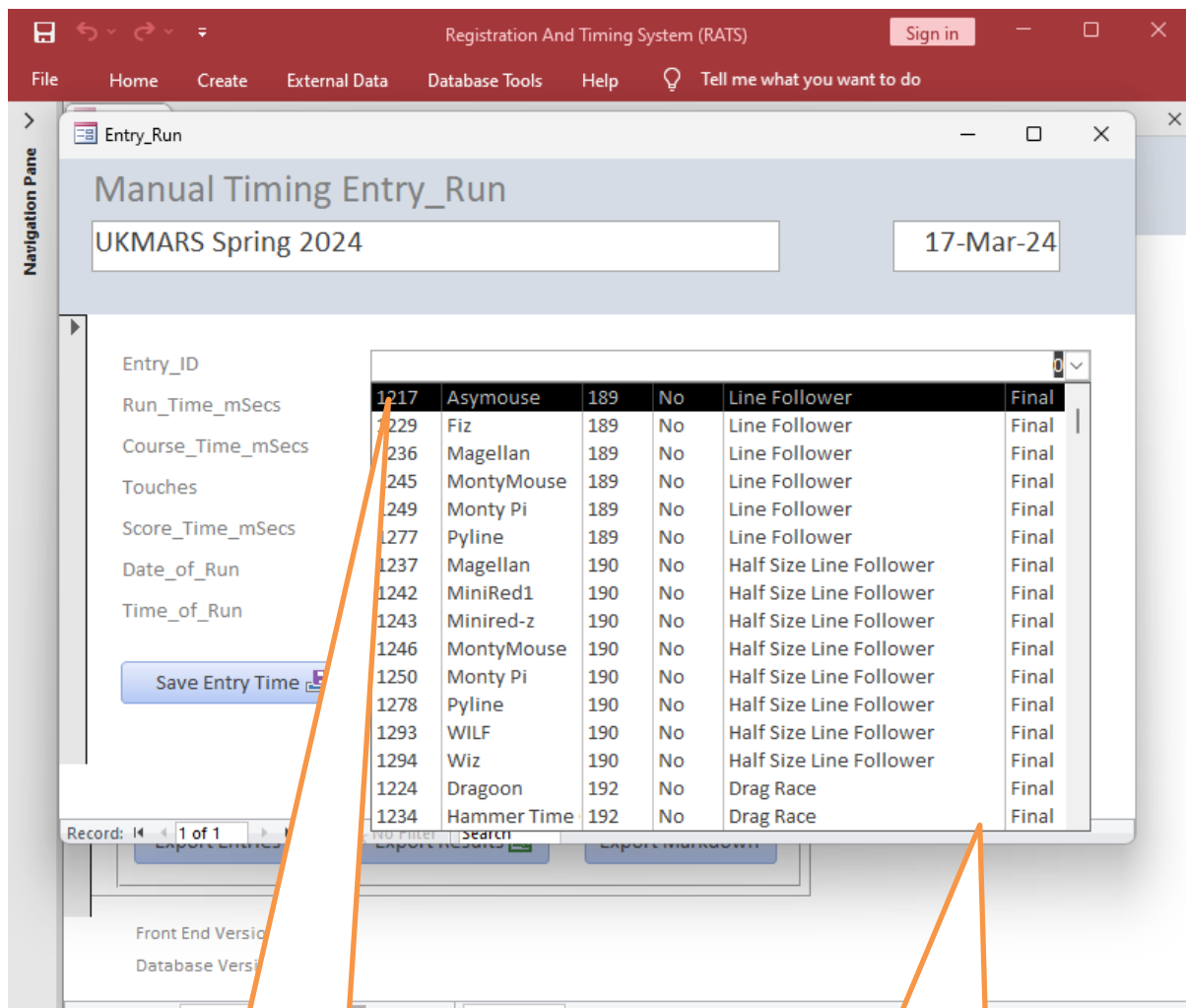
Front End Version 3.7
Database Version 3.3b

Record: 1 of 20 No Filter Search

Form View Num Lock

1. Click to manually enter timing results.

It is possible to manually record timing results on the database but should be avoided if at all possible!



2. Select the entry for which you are recording the result.

1. A list of unused entries for the current Robotics Event is presented.

Registration And Timing System (RATS) Sign in

File Home Create External Data Database Tools Help Tell me what you want to do

Entry_Run

Manual Timing Entry_Run

UKMARS Spring 2024 17-Mar-24

Entry_ID	1217
Run_Time_mSecs	12987
Course_Time_mSecs	12987
Touches	0
Score_Time_mSecs	12987
Date_of_Run	17-Mar-24
Time_of_Run	10:23:29
Manual_Timing	Yes

Save Entry Time Make Best Time

Record: 1 of 1 No Filter Search

Front End Version 3.7 Database Version 3.3b

Record: 1 of 20 No Filter Search

orm View

1. Enter the results for the entry Note the units for time are milliseconds

1. Either save an entry run time that is not the best time or enter the best entry run time

Once the entry has been selected, all times are entered in milliseconds and the date and time of the run can be entered.

If the time is the best score time for the robot the “Make Best Time” button is used and the robot entry will be complete. It is not possible to add further times for this entry once the best time has been recorded.

If the time is a slower run being recorded for completeness use the “Save Entry Time” button, This will record the run time but leave the entry open.

2.13. Post Event Reporting

The screenshot shows the 'Context' page of the Registration And Timing System (RATS). The page has a red header with navigation links: File, Home, Create, External Data, Database Tools, Help, and a 'Tell me what you want to do' search bar. A 'Sign in' button is in the top right. The main content area is titled 'Context' and contains several sections of buttons and form fields.

Callout 1 (top right): 3 Reports suitable for printing. Unsplit scores are not separated between senior and junior. This callout points to the 'Unsplit Scores' button.

Callout 2 (bottom left): 2 Excel format files that are created on the R: drive root folder. This callout points to the 'Export Entries' and 'Export Results' buttons.

Callout 3 (bottom right): Markdown format file suitable for pasting into WordPress that is created on the R: drive root folder. This callout points to the 'Export Markdown' button.

The 'Context' page includes the following elements:

- Event ID: 36 (with 'New Event' button)
- Effective Date: 17/03/2024 (with 'Refresh' button)
- Current Event: UKMARS Spring 2024
- Buttons: Record Entries, Sequence, List Contests, List Ent, Manual Input, Scores Report, Unsplit Scores, All Runs, Export Entries, Export Results, Export Markdown.
- Footer: Front End Version 4.7, Database Version 3.3b, No Filter, Search.

3. Timing Supervisor – RATSts

3.1. Introduction

3.1.1. Windows

RATSts has multiple windows:

- Supervisor window
- Timer display windows
- A separate running order window
- Results window

All the windows are controlled from the Supervisor window

3.1.2. Modes

RATSts has 3 modes:

- Practice Mode
- Contest Mode
- Stand Alone Mode

RATSts normally starts in **Practice Mode** where it is connected to a contest database but timing results are only displayed for the current run and are not stored against robot entries.

In **Contest Mode** (controlled by the supervisor window) the contest class Heat/Final is selected together with the Contest and the Robot, Timing results are stored for the selected robot entry.

Stand Alone Mode is entered automatically if RATSts cannot connect to a contest database. It is the same as practice mode and is useful in allowing timings to be displayed without the need for a database.

3.1.3. Message Log

Every time RATSts is started it creates a message log on the Desktop. The filename is timestamped and the log contains timestamped records for every message sent between RATSts and the timer device and an annotated message everytime a run time is received from the timer showing which robot the time is for.

In the unlikely event of a database error run times can be recovered from this log.

3.2. Timing Supervisor Start Screen

Registration And Timing System (RATS) Version 3.9.1 database version 3.3

17/03/2024 UKMARS Spring 2024

Practice Mode

Selected Contest Class

Current Contestant

Course	To Go	Run No	of	Touces	BestScore	Rank	Last Run	LastScore	SplitTime
seCS	seCS	no	no	no	calc	calc	seCS	calc	seCS

Timing Gates State State
Timing Gates State

Touch Enabled Touch Enabled
Touch Divider Touch Divider

Touch Cumtive Touch Cum

Touch Time Touch Time

Course Divider
Maze Divider

Co Timer: OFF Select Class

Switch Mode Practice <=> Contest

Entry Controls

Add Touch

DNF Clear Display

New Robot

1 Channel Timer

800x600 1280x720 Results

612 x 595 Run Order

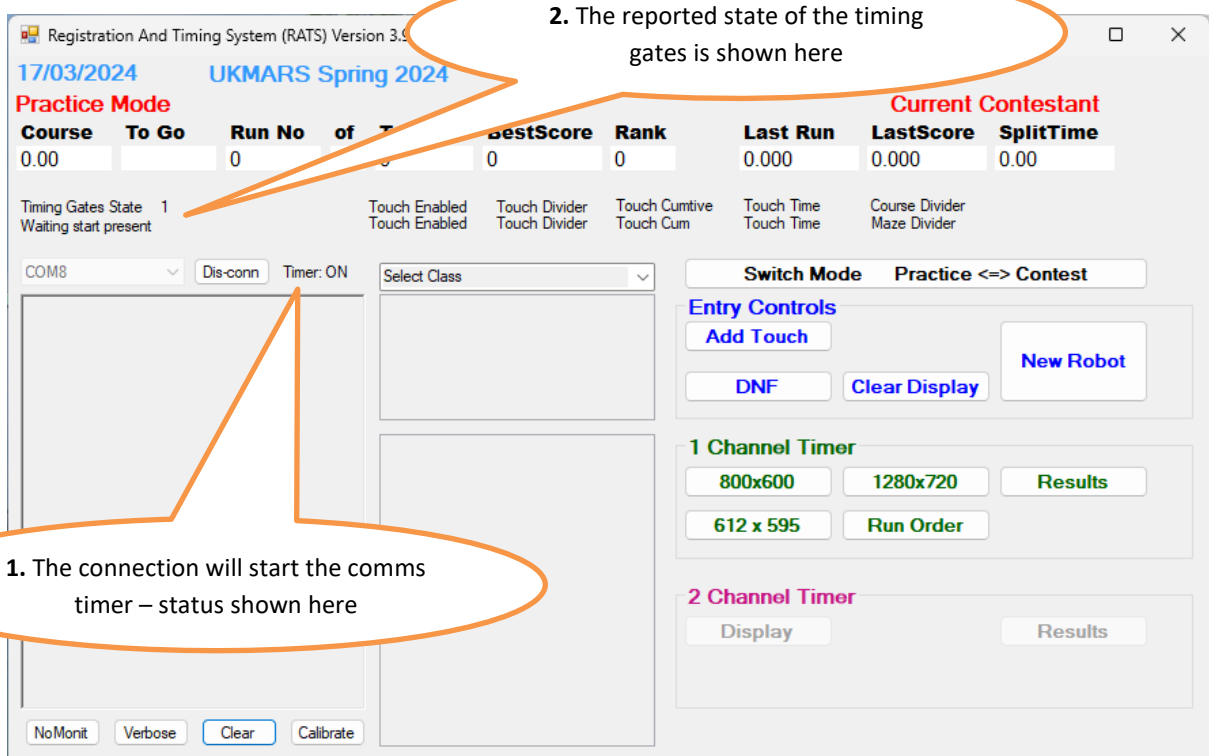
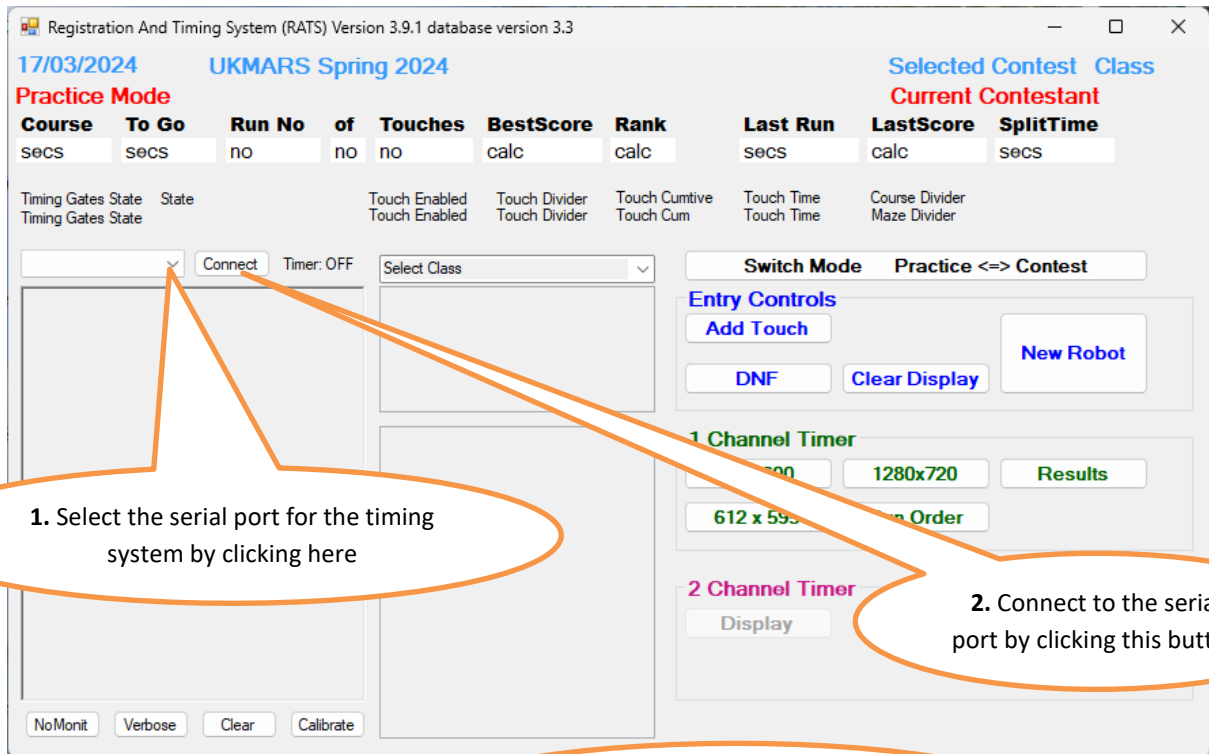
2 Channel Timer

Display Results

NoMonit Verbose Clear Calibrate

1. Date and title of the Robotics Event are automatically filled from RATSdb

Before starting RATSts, connect the timing system to the PC using the USB cable. Allow the connection to complete any driver installation.



3.3. Timing Supervisor Practice Mode

The screenshot shows the 'Registration And Timing System (RATS) Version 3.9.1 database version 3.3' interface. The title bar indicates the date '17/03/2024' and the event 'UKMARS Spring 2024'. The main window is titled 'Practice Mode' and displays a table of contestant data. The table has columns: Course, To Go, Run No, of, Touches, BestScore, Rank, Last Run, LastScore, and SplitTime. The current contestant's data is shown in the first row: Course 27.79, To Go, Run No 0, of no, Touches 0, BestScore 0, Rank 0, Last Run 0.000, LastScore 0.000, and SplitTime 10.53. Below the table, there are various controls including 'Timing Gates State', 'Touch Enabled', 'Touch Divider', 'Touch Cumtive', 'Touch Time', 'Course Divider', and 'Mode'. A 'New Robot' button is highlighted with a callout. Another callout points to the '1 Channel Timer' section, which includes buttons for '800x600', '1280x720', '612x338', and 'Run Order'. A third callout points to the '2 Channel Timer' section, which includes a 'Display' button. A fourth callout points to the 'Controls' section, which includes buttons for 'Add Touch', 'DNF', 'Clear Display', and 'New Robot'.

1. With the maze timing gates connected, the timer will automatically work in practice mode

2. Clicking this button will reset the timer but no times are stored in practice mode

3. Clicking these buttons will open the display window that can be moved to a projector or large screen. Choose the best aspect ratio

The screenshot shows the 'single_ch_display_16_9' window titled 'UKMARS Spring 2024 Practice Mode'. The main display area shows a large digital timer with the time '0:24.863'. Below the timer, there are several data fields: 'Touches' (0), 'Run' (empty), 'Of' (empty), 'Time Left' (empty), 'Best Score' (00:00.00), and 'Rank' (empty). The text 'Run Times' is displayed below the 'Touches' field, and 'Best Score Times for all Robots' is displayed below the 'Best Score' field. On the right side of the window, there is a section titled 'Running Order of Next Entries'.

While contestants are practicing before a competition there is no need to take any action. The timing gates will trigger from the start cell, start gate and finish gate and will display times for contestants to measure their performance.

3.4. Timing Supervisor Competition Mode

Registration And Timing System (RATS) Version 3.9.1 database version 3.3

17/03/2024 UKMARS Spring 2024

Practice Mode

Course	To Go	Run No	of	Touches	BestScore	Rank	Last Run	LastScore	SplitTime
0.00		0	no	0	0	0	0.000	0.000	0.00

Timing Gates State 1
Waiting start present

Touch Enabled Touch Enabled Touch Divider Touch Divider Touch Cumtive Touch Cum Touch Time Touch Time Course Divider Maze Divider

COM8 Dis-conn Timer: ON Select Class

Switch Mode Practice <=> Contest

Entry Controls

Add Touch

DNF Clear Display New Robot

1 Channel Timer

800x600 1280x720 Results

612 x 595 Run Order

2 Channel Timer

Display Results

NoMonit Verbose Clear Calibrate

1. Clicking this button swaps between practice mode and competition mode

Registration And Timing System (RATS) Version 3.9.1 database version 3.3

17/03/2024 UKMARS Spring 2024

Current Contestant

Course	To Go	Run No	of	Touches	BestScore	Rank	Last Run	LastScore	SplitTime
0.00	0.00	0	no	0	0	0	0.000	0.000	0.00

Timing Gates State 1
Waiting start present

Touch Enabled Touch Enabled Touch Divider Touch Divider Touch Cumtive Touch Cum Touch Time Touch Time Course Divider Maze Divider

COM8 Dis-conn Timer: ON Select Class

Switch Mode Practice <=> Contest

Entry Controls

Add Touch

DNF Clear Display New Robot

1 Channel Timer

800x600 1280x720 Re

612 x 595 Run Order

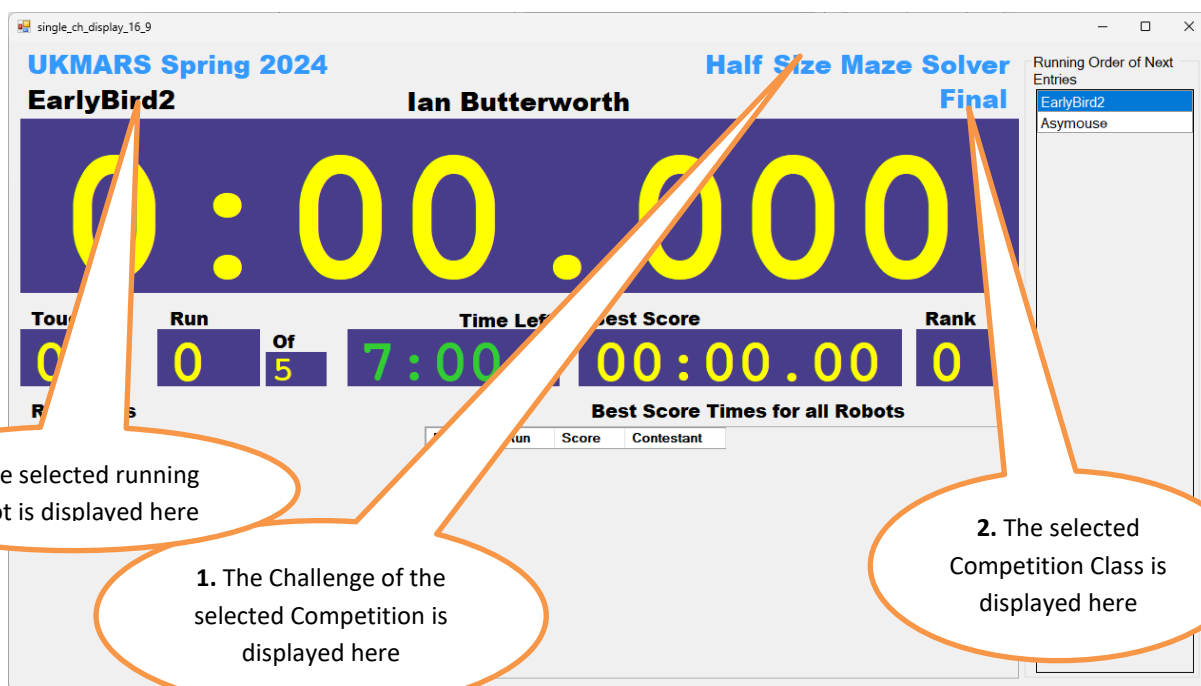
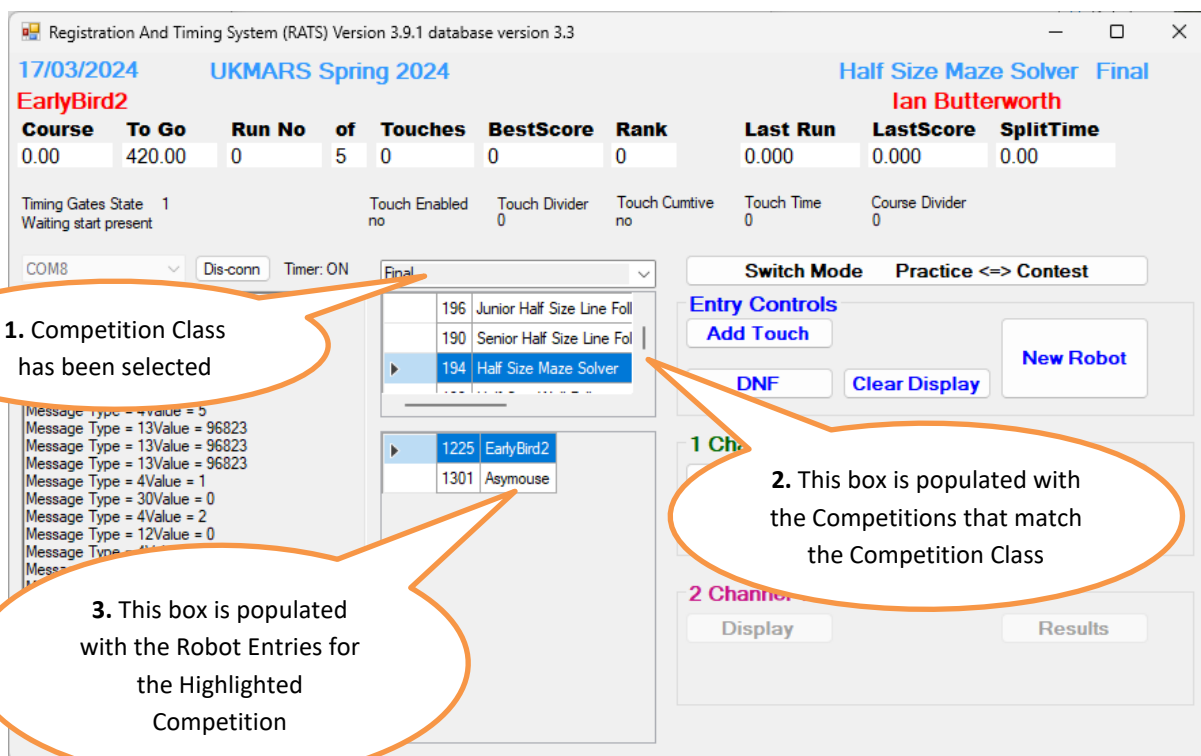
2 Channel Timer

Display Results

NoMonit Verbose Clear Calibrate

2. Click here to select the class of competition to be run, either Final or Heats

1. Click here to reset the timer



Once you have selected the Robot for the Competition in the Competition Class, the Contestant can place their Robot in the starting cell. The timer will automatically start when their Robot triggers the start gate.

UKMARS Spring 2024 Junior Test Robot **Maze Solver Final**

1. Timing starts when Robot triggers the start gate

4. The running order for future entries is shown here

2. All run times for this Robot are shown here

3. The best score times for earlier Robots are shown here

Time: 1:26.739

Touches: 0 **Run:** 1 **Of:** 5 **Time Left:** 5:17 **Best Score:** 1:26.739 **Rank:** 11

Run Times

Time of Run	Run	Touches	Score
13:42:56	86.739	0	86.739

Best Score Times for all Robots

Robot	Run	Score	Contestant
Mazerunner32	7.6	7.6	Peter Harrison
MouseX2	9.314	9.314	Derek Hall
HAL 900	10.456	10.456	Jim Chidley
Mr fusion	10.898	10.898	Derek Hall
Decimus 5A	13.195	13.195	Peter Harrison
PicOne Turbo	19.237	19.237	Jim Chidley
MontyMouse	25.935	25.935	Neil Stevenson
Topaz	58.979	58.979	Bernard Grabowski
R3D3a	66.276	66.276	Malcolm Loggie
PicOne	68.801	68.801	Jim Chidley
Junior TestRobot	86.739	86.739	Junior Test

UKMARS Spring 2024 Junior Test Robot **Maze Solver Final**

1. The time remaining in seconds for this entry is displayed here

2. The best score time for this robot displayed here

3. The current position of this robot in the ranking table is displayed here

Time: 1:26.739

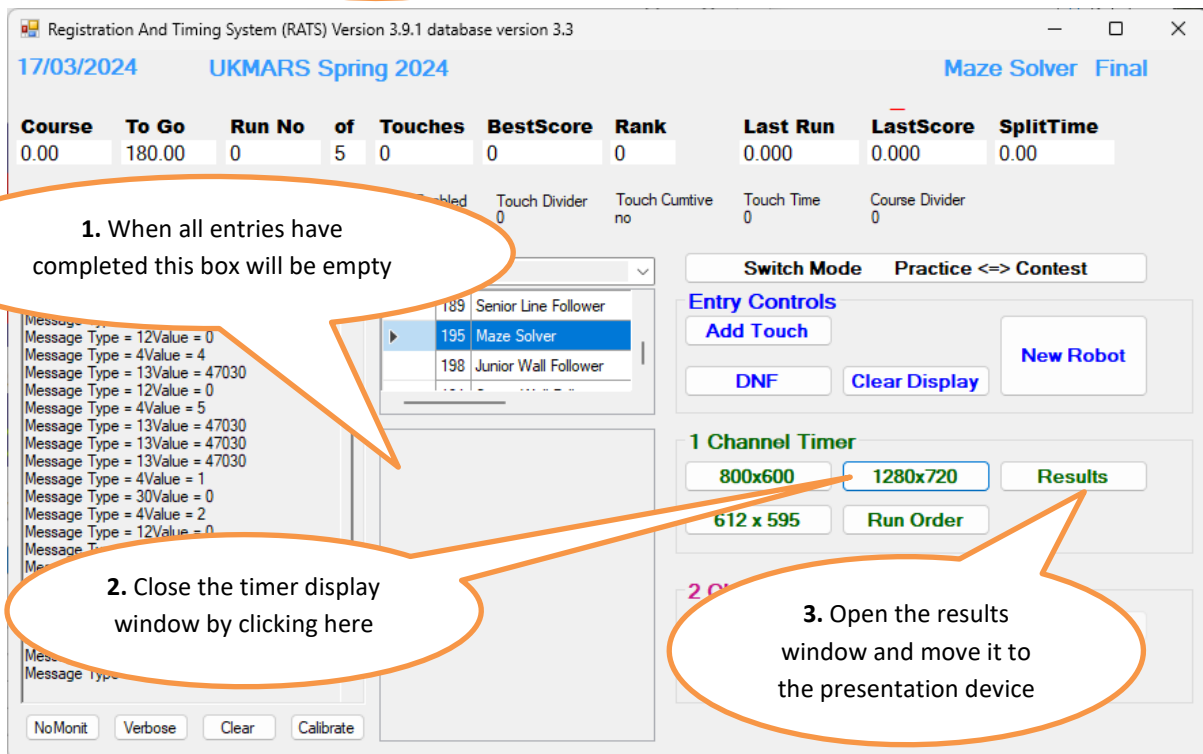
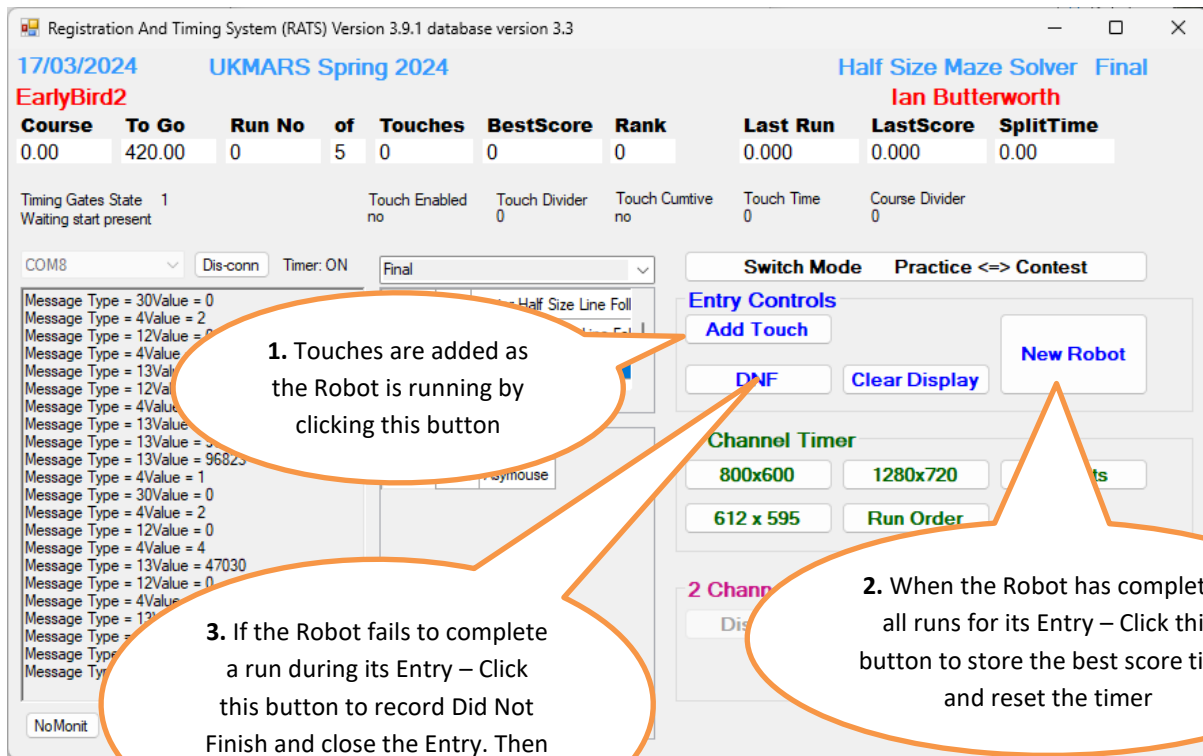
Touches: 0 **Run:** 1 **Of:** 5 **Time Left:** 5:17 **Best Score:** 1:26.739 **Rank:** 11

Run Times

Time of Run	Run	Touches	Score
13:42:56	86.739	0	86.739

Best Score Times for all Robots

Robot	Run	Score	Contestant
Mazerunner32	7.6	7.6	Peter Harrison
MouseX2	9.314	9.314	Derek Hall
HAL 900	10.456	10.456	Jim Chidley
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MontyMouse	25.935	25.935	Neil Stevenson
Topaz	58.979	58.979	Bernard Grabowski
R3D3a	66.276	66.276	Malcolm Loggie
PicOne	68.801	68.801	Jim Chidley
Junior TestRobot	86.739	86.739	Junior Test



Results Maze Solver

Final

Robot	Run Time	Score Time	Contestant
Mazerunner32	7.6	7.6	Peter Harrison
MouseX2	9.314	9.314	Derek Hall
HAL 900	10.456	10.456	Jim Chidley
Mr fusion	10.898	10.898	Derek Hall
Decimus 5A	13.195	13.195	Peter Harrison
PicOne Turbo	19.237	19.237	Jim Chidley
MontyMouse	25.935	25.935	Neil Stevenson
Topaz	58.979	58.979	Bernard Grabowski
R3D3a	66.276	66.276	Malcolm Loggie
PicOne	68.801	68.801	Jim Chidley
Junior Test Robot	86.739	86.739	Junior Test

3.5. Timing Supervisor Setup Aids

Registration And Timing System (RATS) Version 3.9.1 database version 3.3

17/03/2024 UKMARS Spring 2024

Practice Mode

Course	To Go	Run No	of	Touches	Current Contestant	Score	SplitTime
0.00		0	no	0		0.00	

Timing Gates State 1 Waiting start present

Touch Enabled Touch Enabled

COM8 Dis-conn Timer: ON

Practice <=> Contest

Controls

Add Touch DNF Clear Display New Robot

0x720 Results Run Order

2 Channel Timing Display

NoMonit Verbose Clear Calibrate

1. The messages passed between RATs and the timing gates are displayed here
2. These 3 buttons control the level of detail displayed
3. If the timing gates support calibration messages, calibration can be started here

Registration And Timing System (RATS) Version 3.9.1 database version 3.3

17/03/2024 UKMARS Spring 2024

Maze Solver Final

Course	To Go	Run No	of	Touches	BestScore	Rank	Last Run	LastScore	SplitTime
0.00	180.00	0	5	0	0	0	0.000	0.000	0.00

Timing Gates State 0 Gates calibrating

Touch Enabled no Touch Divider 0 Touch Cumtve no Touch Time 0 Course Divider 0

COM8 Dis-conn

Calibration

	Pot Value	Brightness Value	Trigger
Start Cell	0	0	True
Start Gate	0	0	
Finish Gate	0	0	

Message Type = 73Value = 1
Message Type = 71Value = 1
Message Type = 72Value = 1
Message Type = 85Value = 0
Message Type = 81Value = 0
Message Type = 83Value = 0
Message Type = 73Value = 1
Message Type = 71Value = 1
Message Type = 72Value = 1
Message Type = 85Value = 0
Message Type = 81Value = 0
Message Type = 83Value = 0
Message Type = 73Value = 1
Message Type = 71Value = 1
Message Type = 72Value = 1
Message Type = 85Value = 0
Message Type = 81Value = 0
Message Type = 83Value = 0
Message Type = 73Value = 1
Message Type = 71Value = 1
Message Type = 72Value = 1
Message Type = 85Value = 0

NoMonit Verbose Clear StopCali

1. Click here to close the calibration window the click NEW ROBOT to reset the timer

Appendix A

RATSdb Entity Relationship Diagram

Relationships for Competitions

Monday, March 26, 2018

